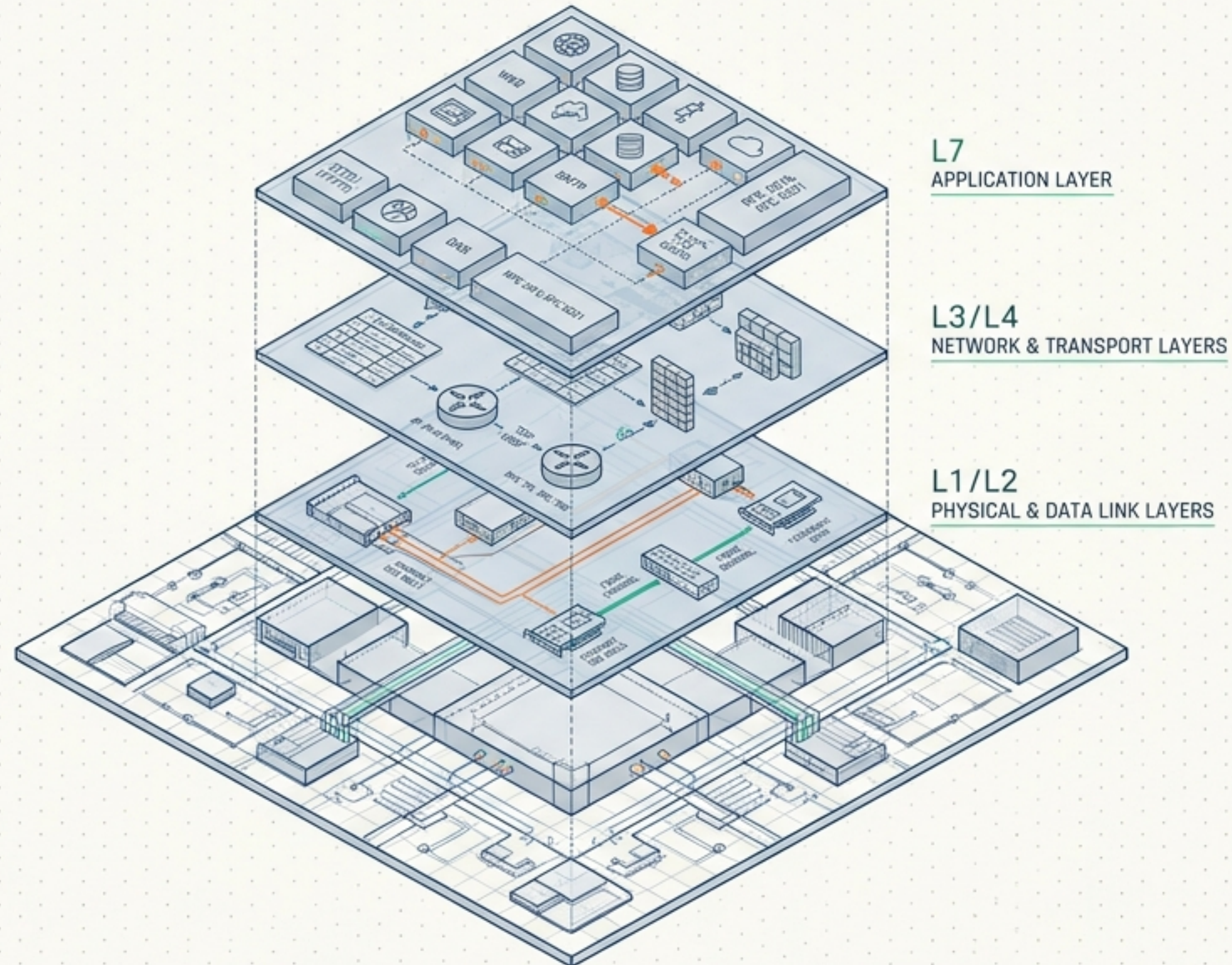


THE BLUEPRINT OF THE INTERNET

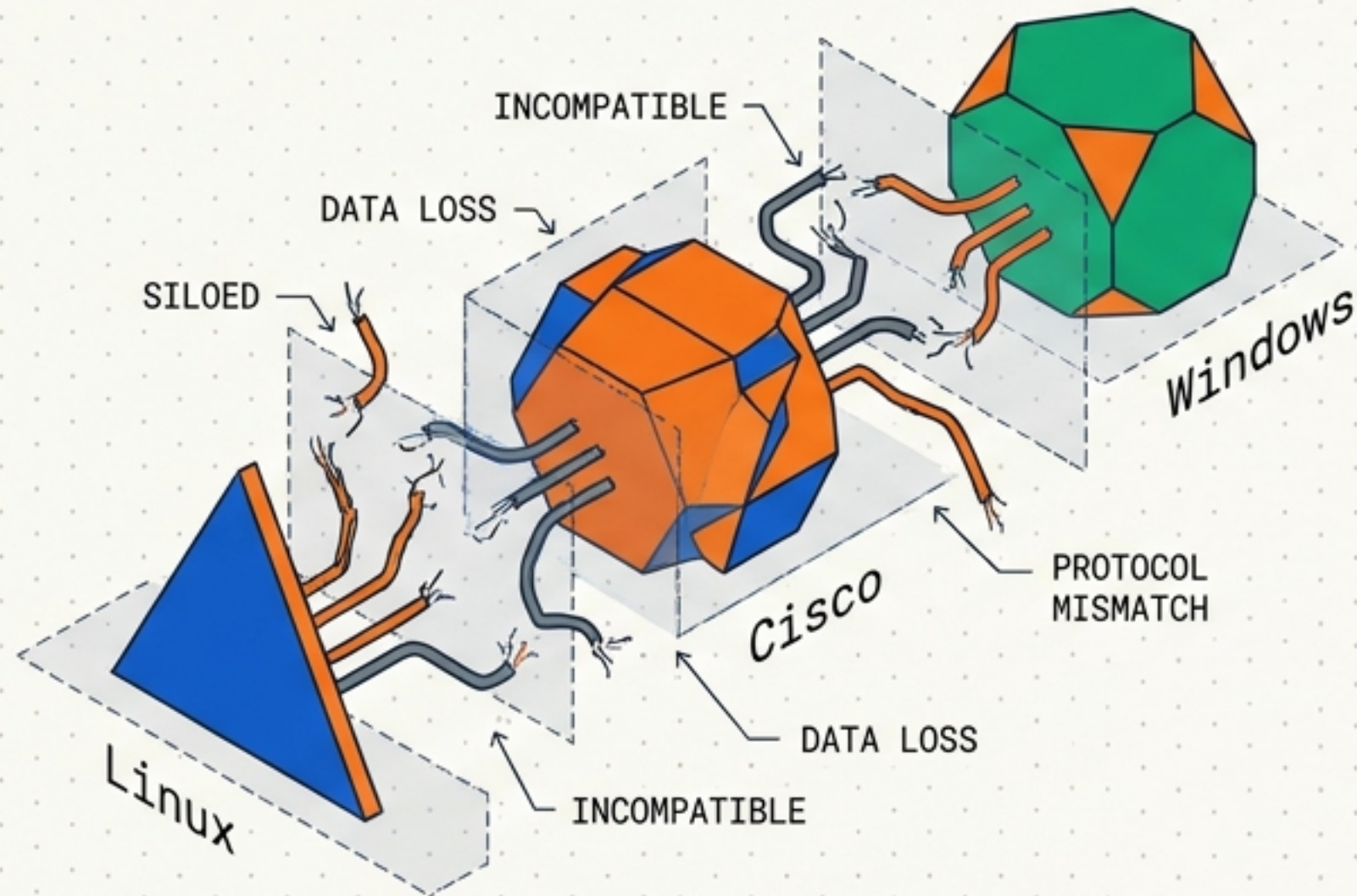
Protocols, RFCs, and the Standards Bodies That Build the Web



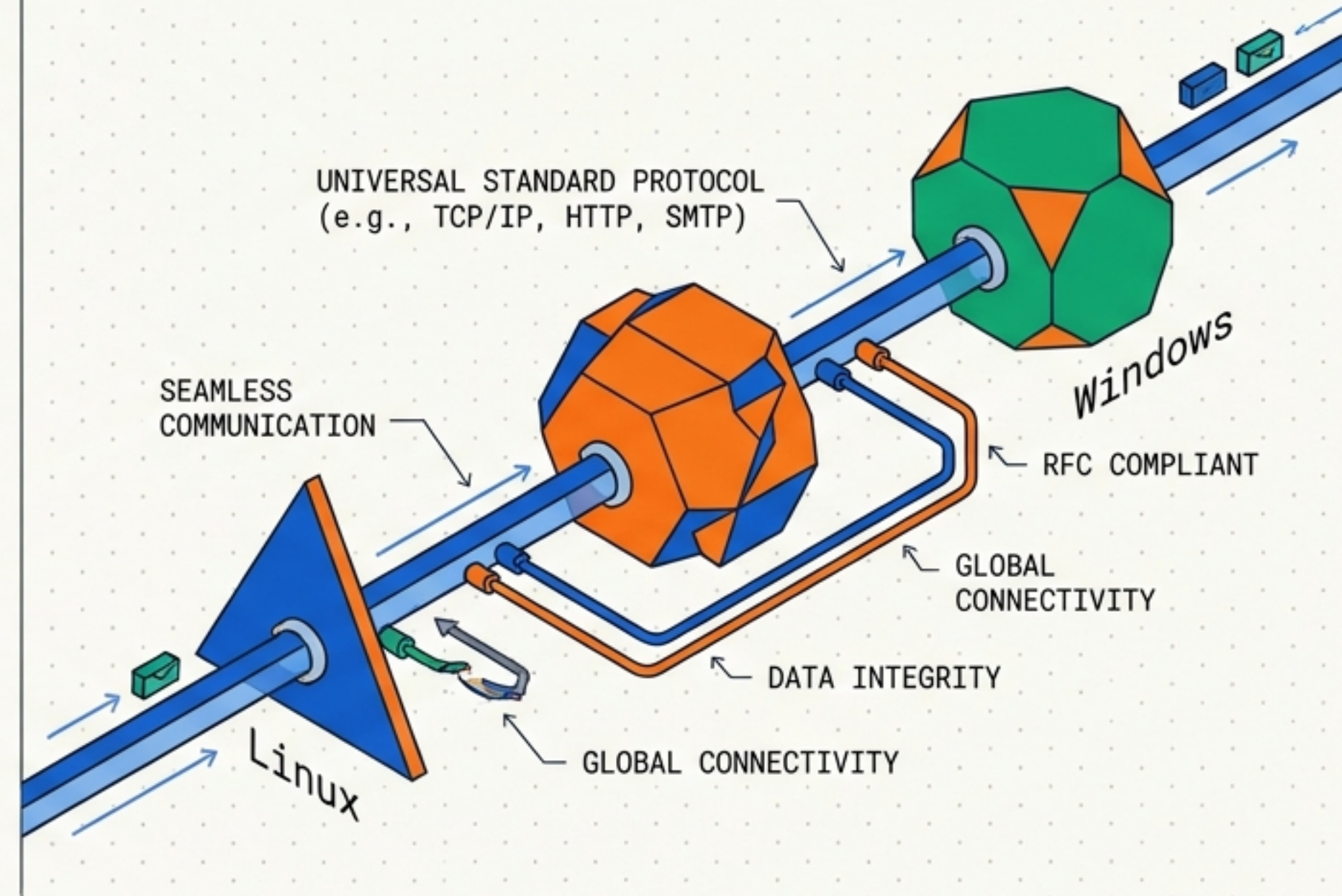
The Threat of Fragmentation

Without standardized protocols, the internet fractures into isolated, proprietary islands. Protocols are the universal language that prevents fragmentation.

Proprietary Chaos



Standardized Interoperability

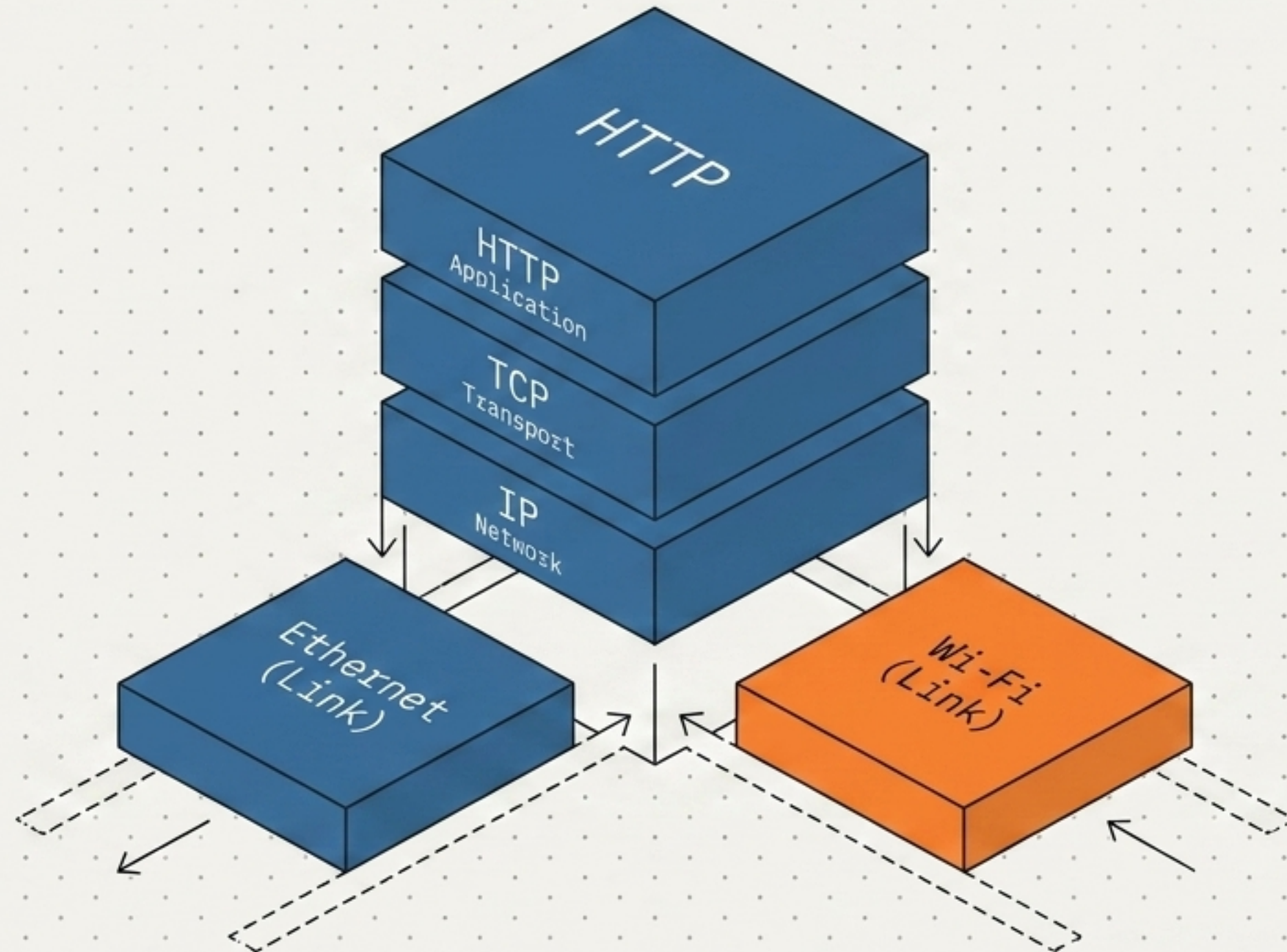


The Three Absolute Rules of a Protocol



Layering and Modular Independence

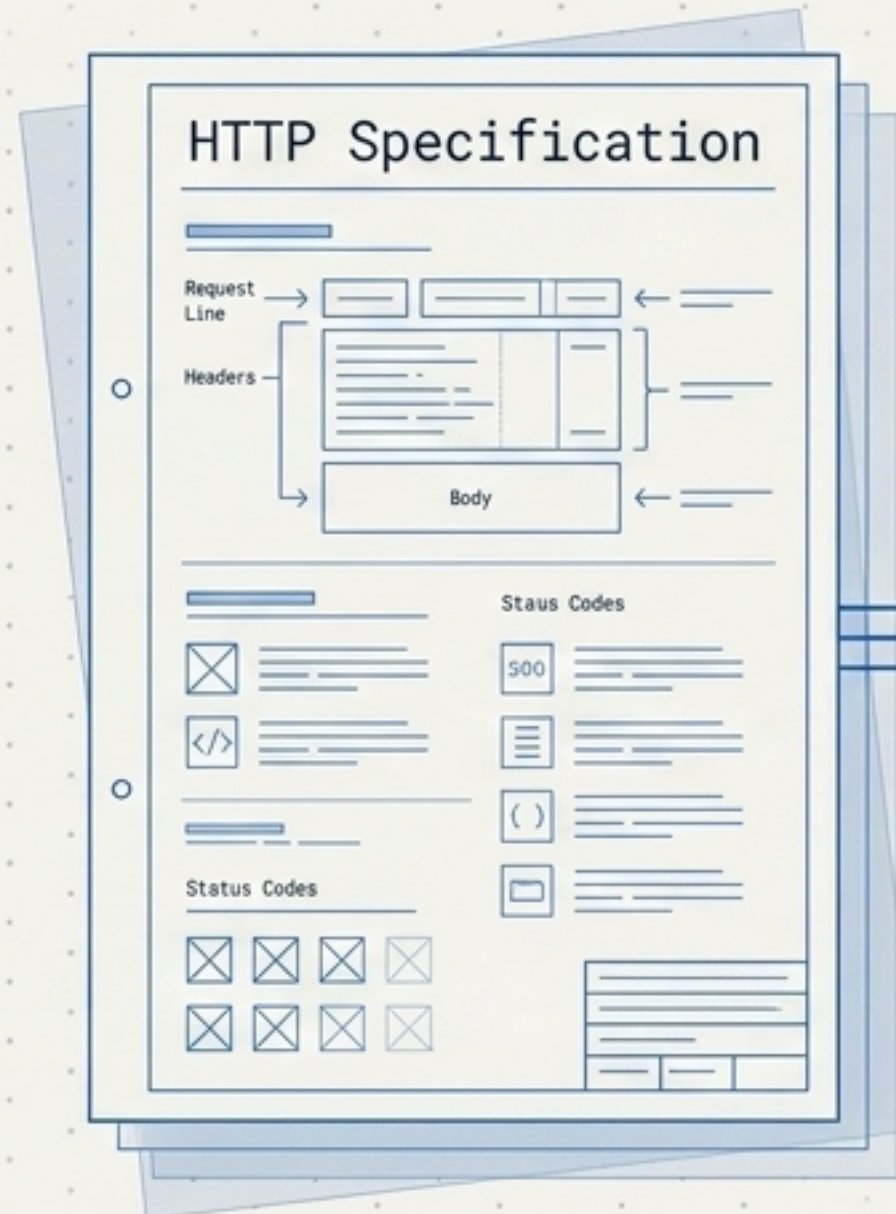
Each protocol focuses on one single job. Because of this layered architecture, a web browser behaves identically whether the physical connection is Ethernet or Wi-Fi.



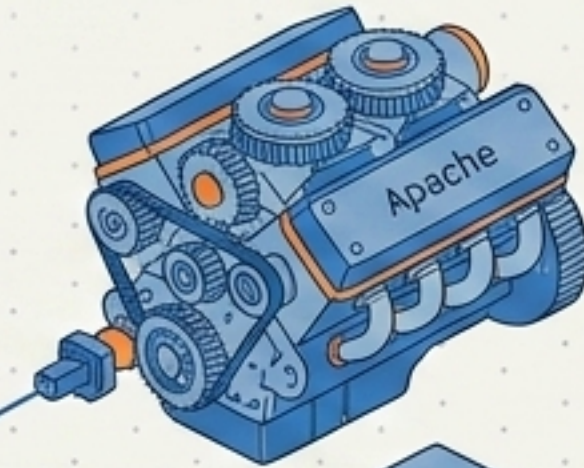
Protocol vs. Implementation

Protocols define the behavior. Implementations execute that behavior.
The source code is different, but the resulting communication is identical.

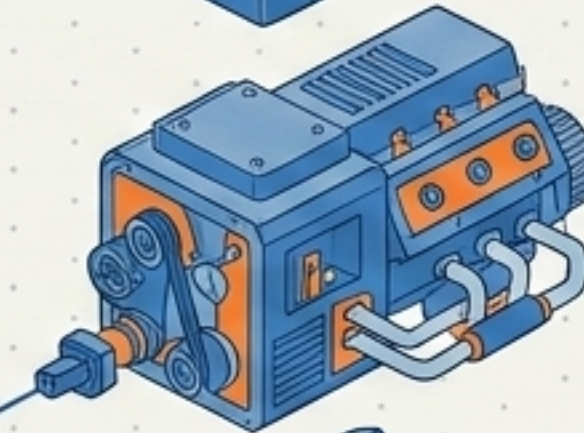
The Specification



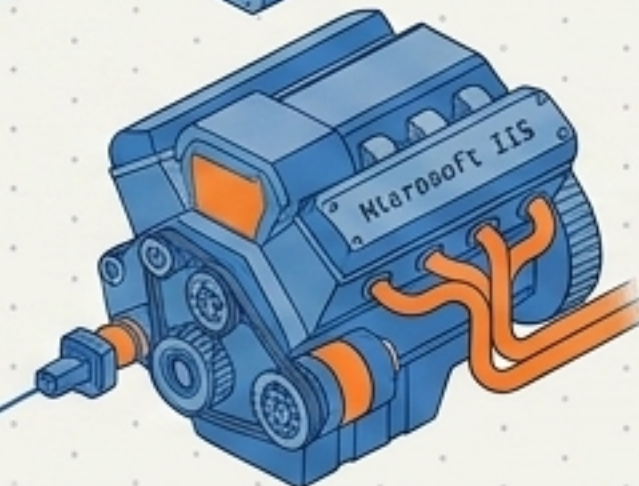
The Implementations



Apache



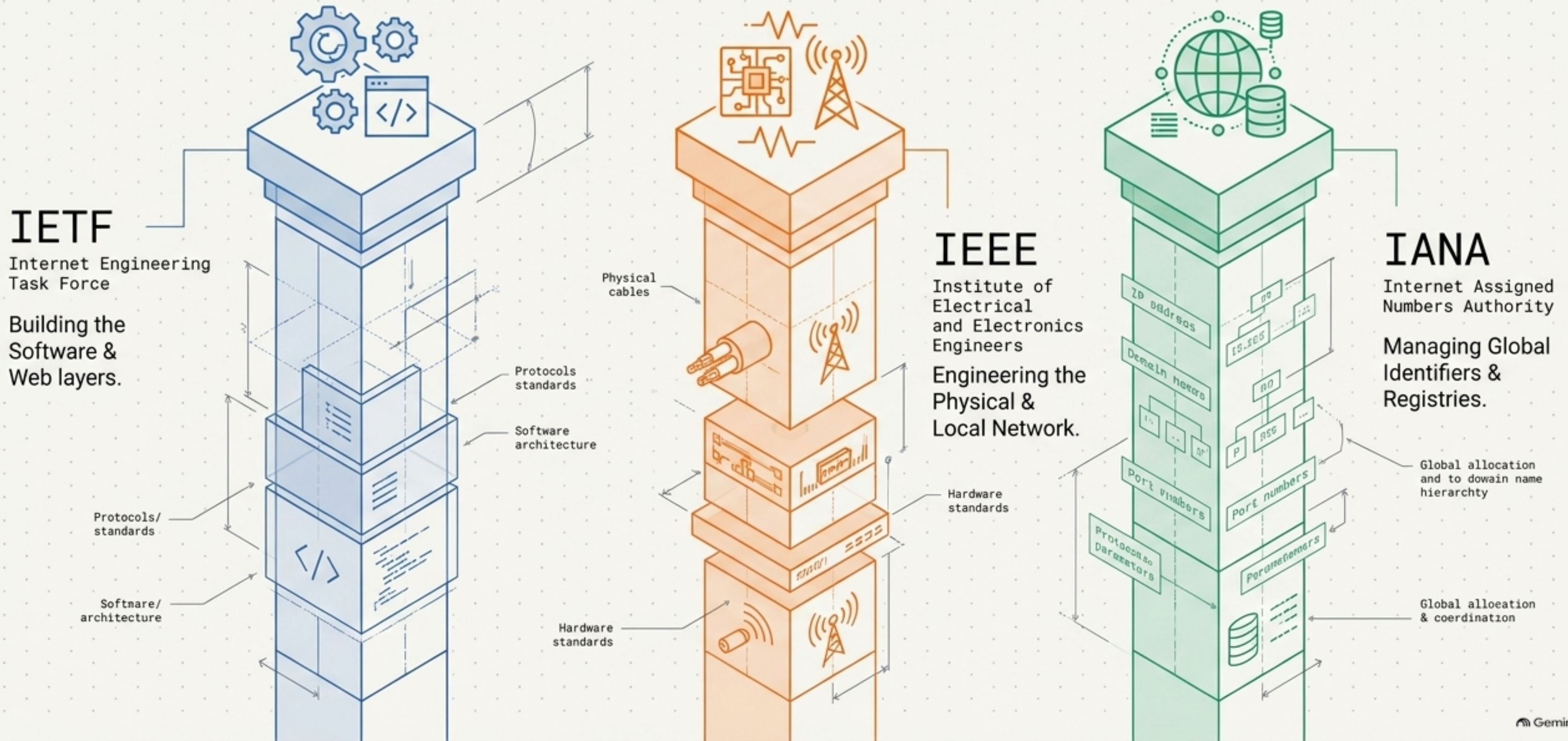
Nginx



Microsoft IIS

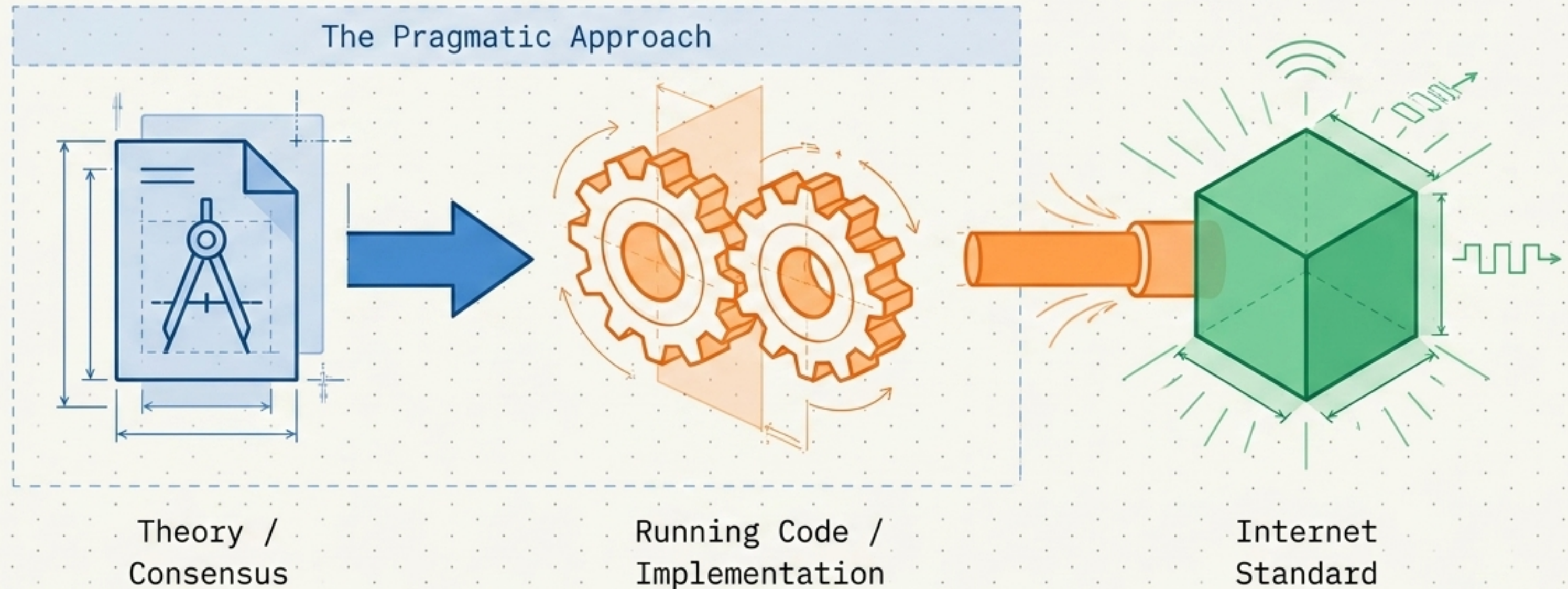
The Architects of the Global Network

The internet is not owned by a corporation; it is governed by open communities setting voluntary technical standards.

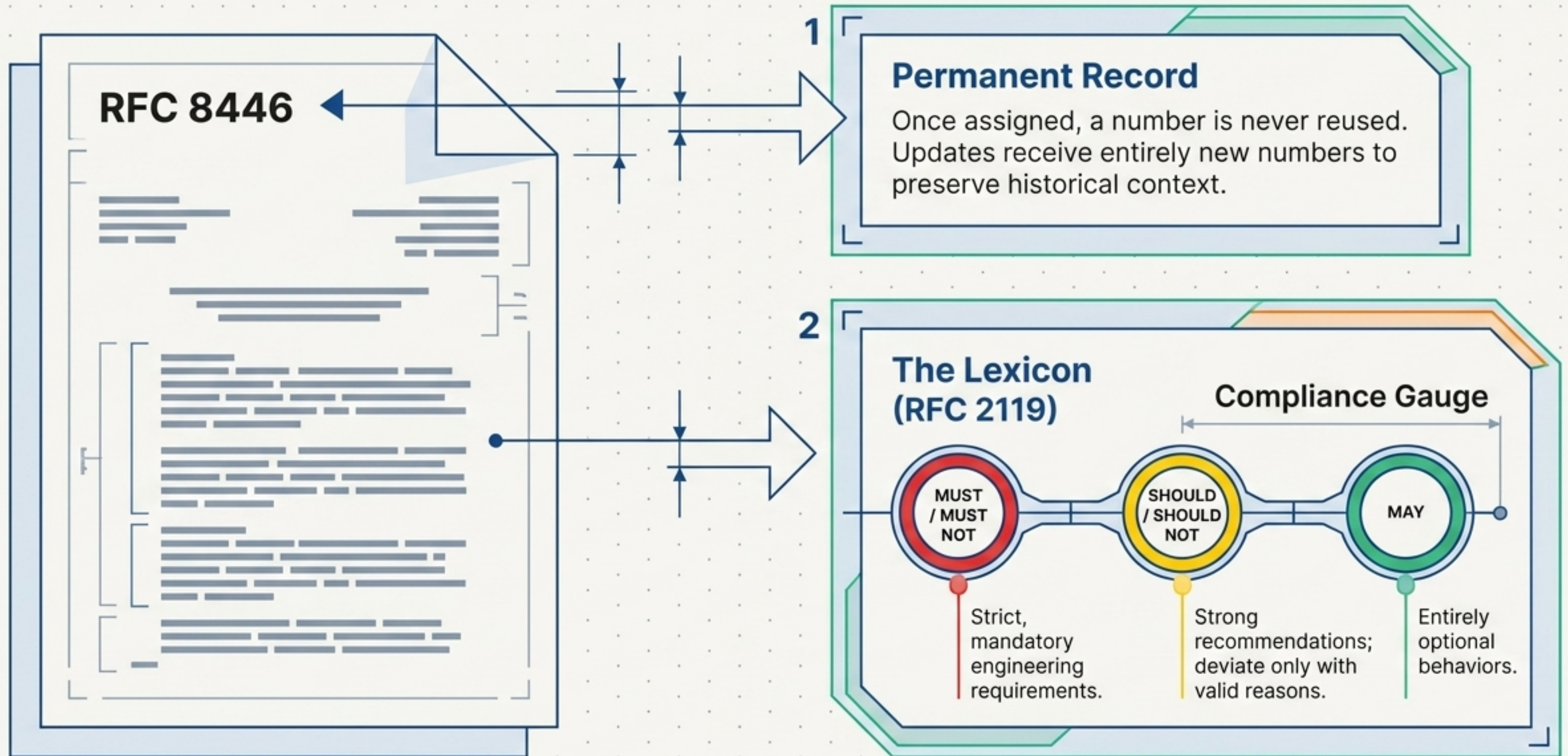


IETF: Rough Consensus and Running Code

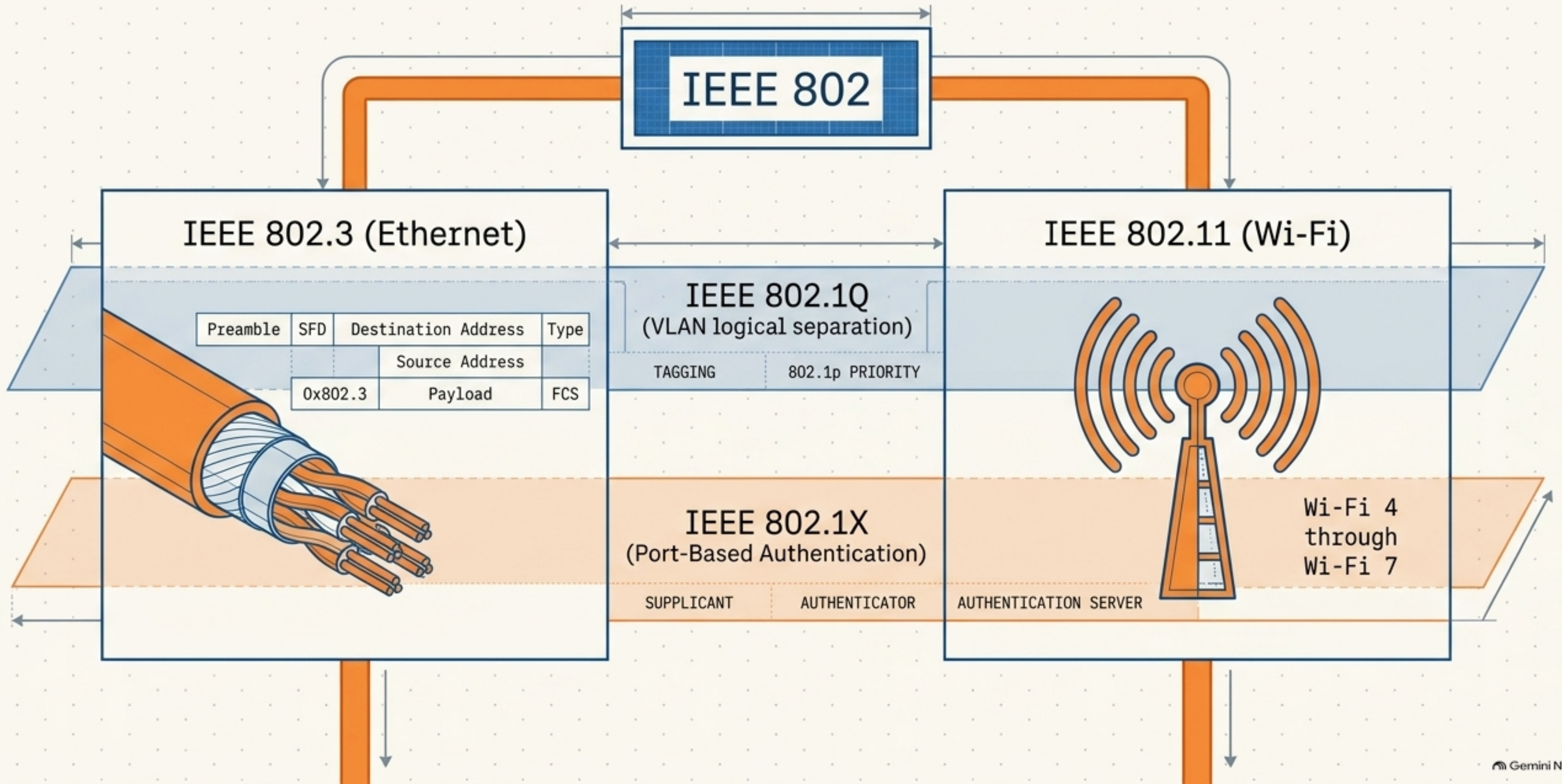
The IETF develops open, interoperable standards. Standards must be proven through working implementations in real-world networks, not just design documents.



The Anatomy of an RFC (Request for Comments)



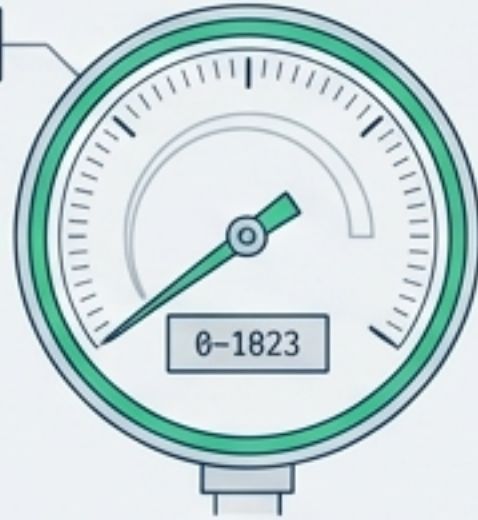
IEEE 802: The Physical Foundation



IANA: The Global Registry

The Dials (Port Numbers)

1



Well-Known Ports (0-1023)

Locked in (e.g., 80 HTTP, 443 HTTPS, 22 SSH).

2



Registered Ports (1024-49151)

Assigned for specific apps.

3

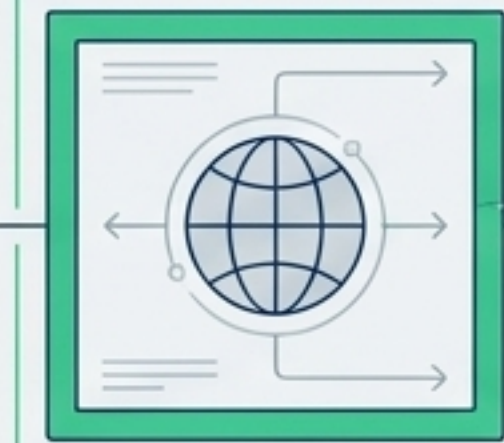


Dynamic Ports (49152-65535)

Temporary, OS-assigned slots.

The Slots (IP & Protocols)

Global IP address blocks routed to Regional Internet Registries (RIRs)



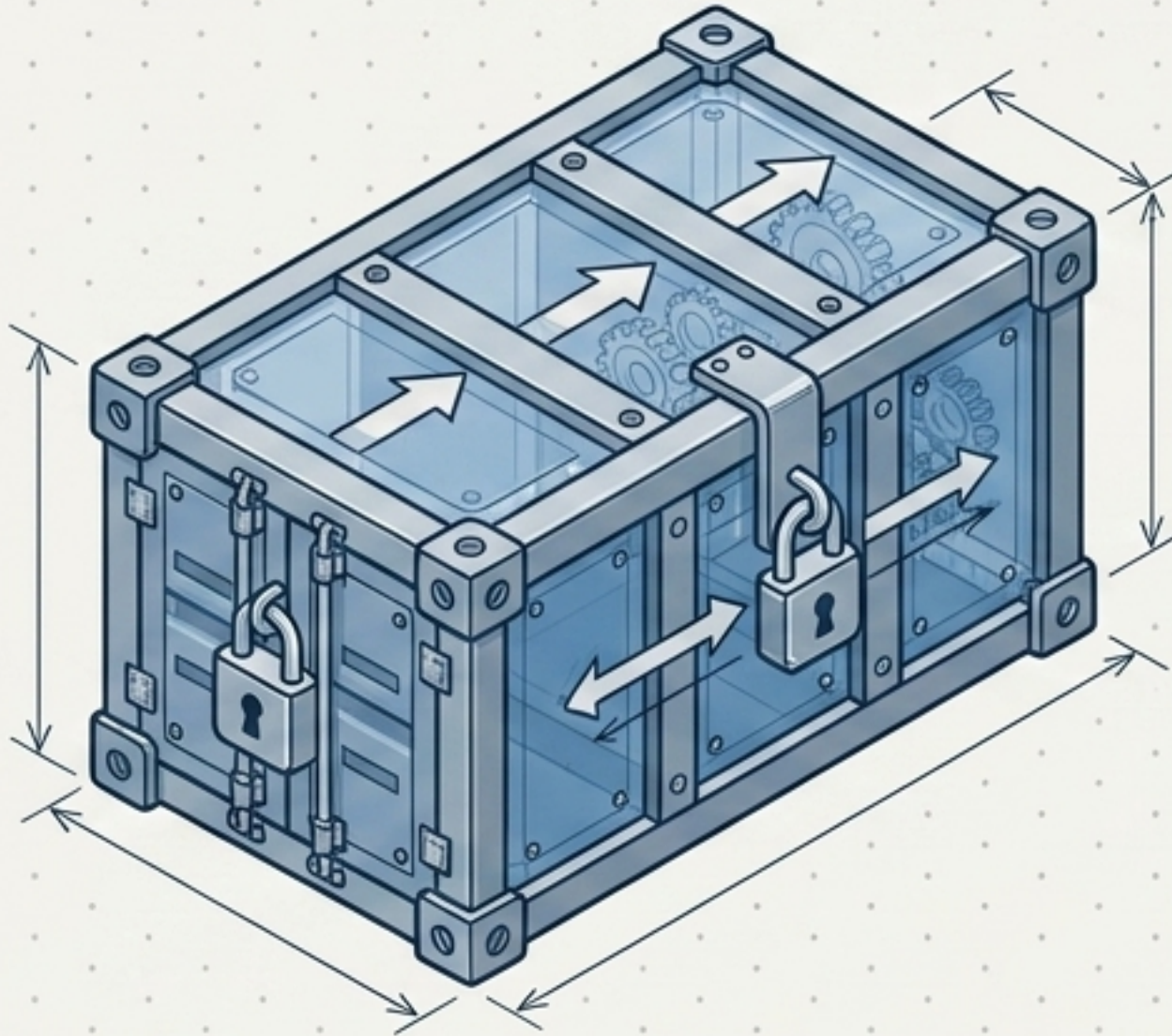
IP Protocol Numbers defining the transport layer payload

The Standards Ecosystem Diagnostic

	IETF	IEEE	IANA
Domain	Internet & Software	Physical LAN & Hardware	Global Registries & Identifiers
Key Output	RFC Specifications	802 Standards	Unique Number Assignments
Example Tech	TCP, HTTP, TLS	Ethernet, Wi-Fi	Port 443, Public IP Blocks

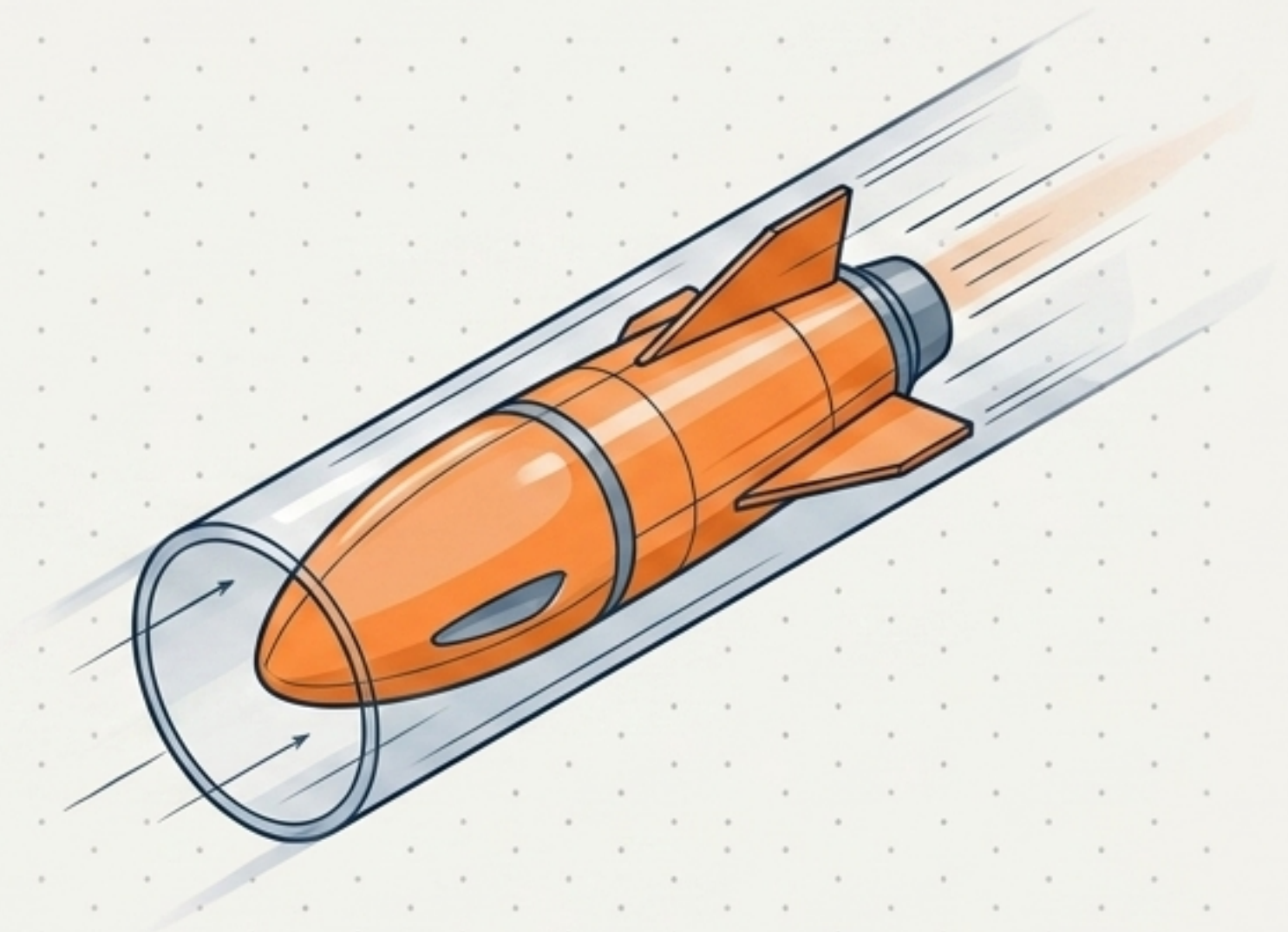
Transport Layer Architecture: TCP vs. UDP

TCP (RFC 793)



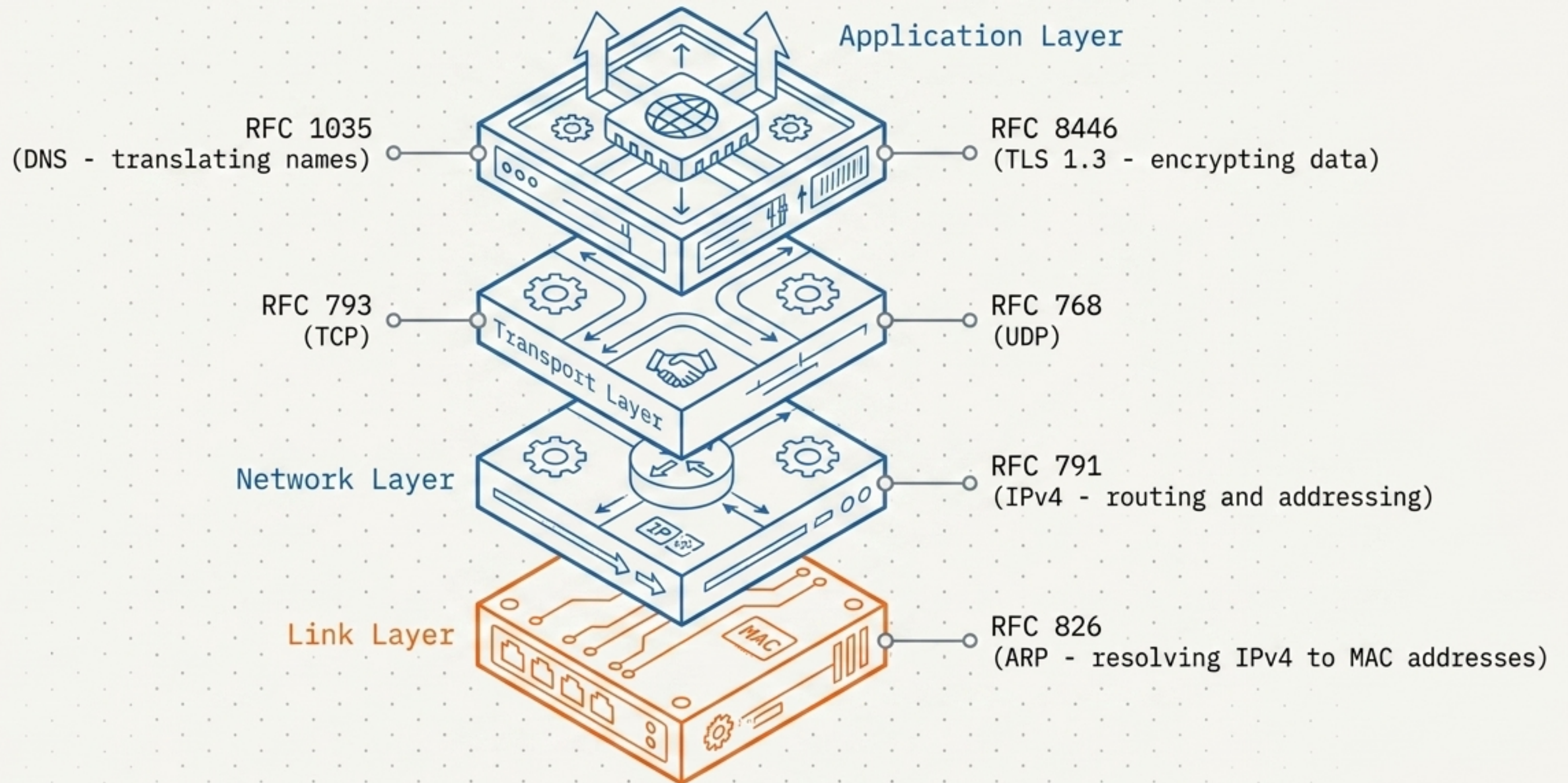
- Reliable & Ordered
- Connection-Oriented (3-Way Handshake)
- Used for: HTTP, SSH, FTP

UDP (RFC 768)



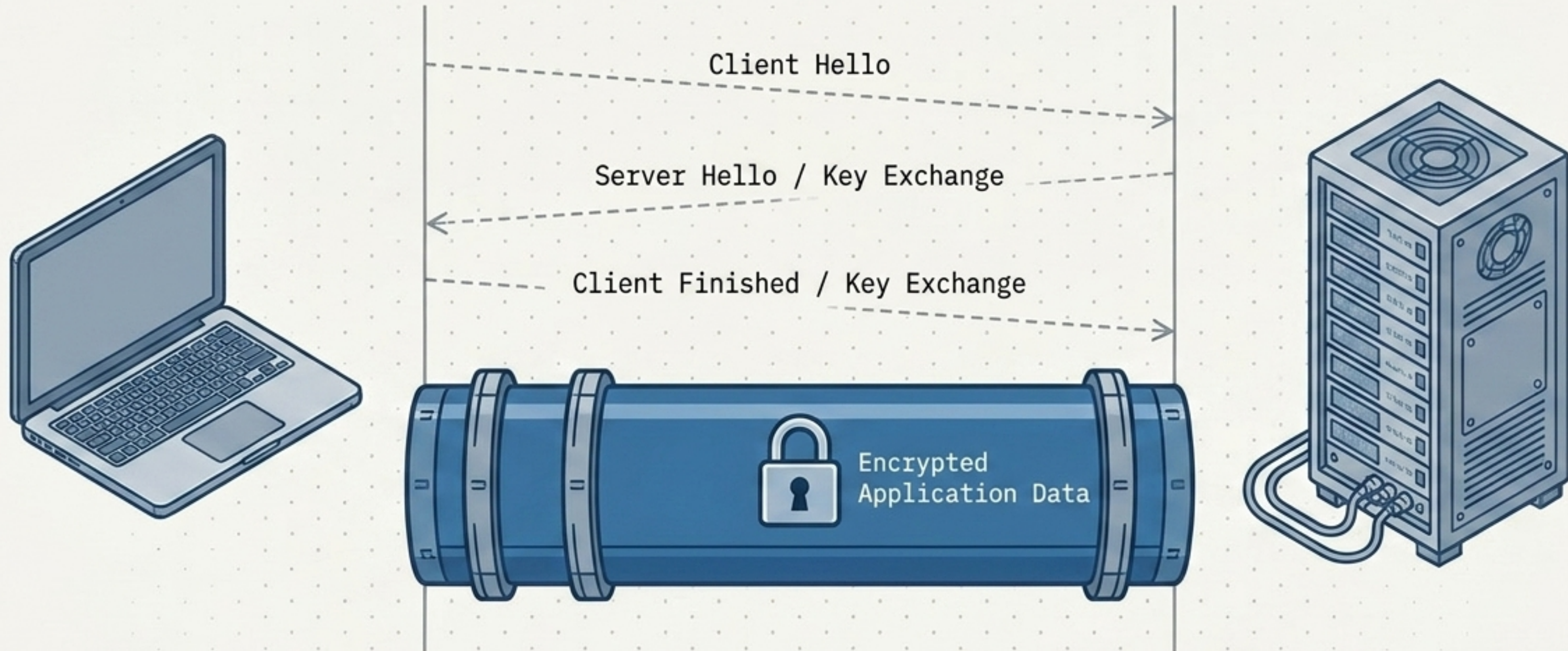
- Fast & Connectionless
- No Delivery Guarantees, Low Overhead
- Used for: DNS, VoIP, Streaming

The Master Map: Foundational RFCs



The Security Layer: RFC 8446 (TLS 1.3)

Modern web traffic relies on TLS to provide Confidentiality, Integrity, and Authentication before any HTTP application data is transmitted.



A Unified Ecosystem in a Single Request

A perfect symphony of standardized rules executing in milliseconds—the true triumph of internet engineering.

